Electronic observables at scale from learned non-orthogonal Hamiltonians

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Abstract

Machine-learned electronic-structure models can generate realistic Hamiltonians for systems containing millions of atoms, but converting these operators into physical observables remains computationally challenging. Here we introduce RSATB, a diagonalization-free framework for general observable extraction from non-orthogonal LCAO Hamiltonians. First, RSATB constructs a reusable graph-local inverse-overlap factor that substantially accelerates repeated metric operations during stochastic propagation. Second, it implements the physical velocity operator in a sparse, periodic, metric-consistent representation. Combined with stochastic Chebyshev evaluation and distributed sparse linear algebra, it provides a unified approach to electronic spectra, real-space densities, local spectroscopy and linear-response observables. Across diamond, silicon and aluminium, RSATB reproduces diagonalization references with errors below 1% for total DOS and below 10% across all observables. Measured computation follows $N^{1.01}$ scaling to 1,004,760 atoms (1.31×10^7 orbitals). Applied to a learned 5.92-million-orbital Hamiltonian of a 455,140-atom twisted-trilayer-graphene supermoiré, RSATB resolves the energy-dependent redistribution of flat-band spectral weight, linking atomistic reconstruction to spatially resolved spectroscopy. Together, these advances bridge large-scale learned Hamiltonians and experimentally relevant observables, extending first-principles-informed modelling to realistic structural complexity.

Machine learning is transforming electronic-structure calculations by enabling the prediction of quantum-mechanical Hamiltonians directly from atomic configurations (Fig. 1) [1–4]. Trained on first-principles data, these models can generate electronic operators for systems containing hundreds of thousands or even millions of atoms, extending atomistic electronic modelling to structural length scales that were previously inaccessible[5, 6]. This development offers an opportunity to connect quantum-mechanical descriptions of electrons with experimentally observed phenomena in structurally complex materials, including disorder, extended defects, interfaces and reconstructed moiré superlattices. Yet a fundamental computational reality remains: the ability to predict an electronic Hamiltonian at scale has advanced far beyond our ability to extract physical observables from it. Without an efficient route from learned Hamiltonians to measurable electronic properties, much of the potential of large-scale machine-learned electronic structure remains unrealized.

The conventional evaluation of electronic observables relies on solving a generalized eigenvalue problem $HC = SC\Lambda$, where H and S are the Hamiltonian and overlap matrices in a localized atomic-orbital basis. The cubic computational complexity and quadratic memory requirements associated with dense diagonalization make this approach impractical for systems containing millions of orbitals[7–9]. Stochastic trace estimators, kernel polynomial expansions and real-time propagation provide an alternative, enabling million-atom calculations of spectra and optical response without explicitly computing individual eigenstates. Such approaches have enabled large-scale calculations using orthogonal tight-binding representations, demonstrating that electronic observables can, in principle, be obtained with computational effort proportional to system size [8–19]. However, most learned Hamiltonians are expressed in non-orthogonal LCAO bases, introducing challenges that are absent or greatly simplified in orthogonal models.

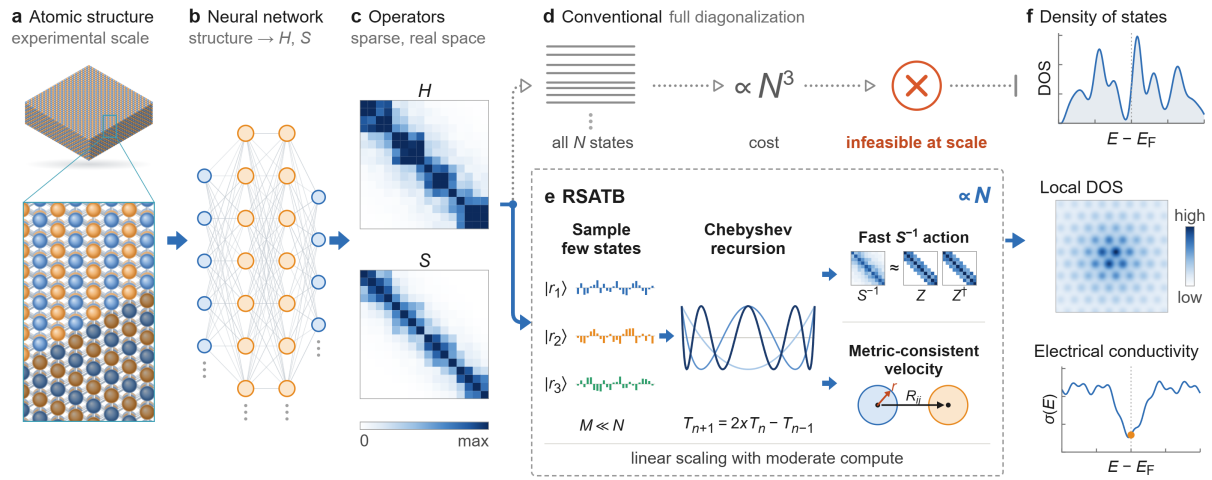
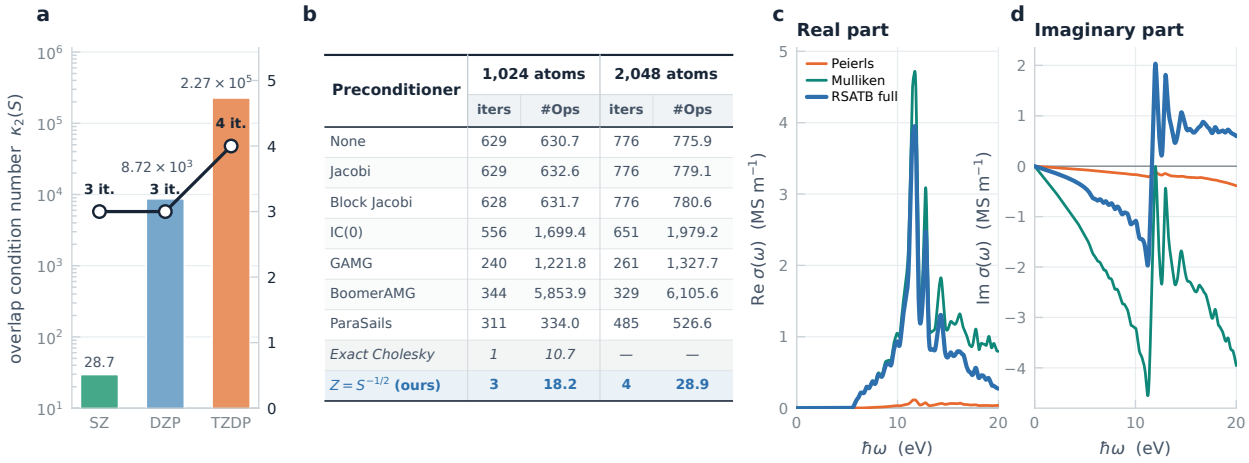


Figure 1: **Schematic of the RSATB workflow.** Neural networks map atomic structures to non-orthogonal electronic operators H and S. Instead of the $O(N_{\mathrm{orb}}^3)$ diagonalization required to recover all eigenstates, RSATB samples a small number of random states and propagates them through Chebyshev recursions, enabling $O(N_{\mathrm{orb}})$ evaluation of physical observables such as DOS, LDOS, and transport response.

In particular, the overlap matrix defines a nontrivial metric, requiring repeated inverse-overlap applications or equivalent orthogonalization during stochastic propagation. Existing approaches, including non-orthogonal kernel polynomial methods, stochastic density-functional formulations, and localized inverse factorization, provide different routes to handling this metric[12–14, 20–22]. However, for extended atomic-orbital bases, its near-linear dependencies can make applying the inverse overlap cost substantially more than applying the Hamiltonian itself, erasing the practical advantage of stochastic evaluation. Recent random Green's-function and chromatic superposition-state methods have addressed density-matrix evaluation, electronic self-consistency, and selected quantum-transport quantities[23–25]. Although distinct from machine-learned Hamiltonian prediction, these approaches similarly extend the scale of accessible electronic-structure calculations, leaving the general observable extraction as a separate computational challenge. Moreover, general response like optical and transport properties pose a further difficulty, requiring physically consistent velocity and current operators that account for both the overlap metric and intra-orbital position contributions[26–29]. Integrating these requirements into a unified, sparse, accurate, and effectively linear-scaling framework for extracting spectral, spatially resolved, and response properties from large learned LCAO Hamiltonians remains an open challenge.

Here we introduce Random-State Ab-initio Tight-Binding (RSATB), a diagonalization-free computational framework that makes spectral, real-space and linear-response observables accessible from sparse, non-orthogonal LCAO Hamiltonians at unprecedented scale. RSATB combines stochastic Chebyshev evaluation with two developments designed specifically to overcome the limitations of non-orthogonal electronic representations. First, we construct a reusable, graph-local approximation to the inverse square root of the overlap matrix, using subgraph localized Krylov approximations and sparsity-aware refinement. This factor provides an efficient treatment of the overlap metric across repeated Hamiltonian operations, drastically reducing the cost of iterative overlap solves. Second, we formulate and implement the periodic velocity operator in a sparse, metric-consistent representation that respects intra-orbital charge displacement. Together with distributed variable-block sparse linear algebra, these developments enable the evaluation of electronic spectra, occupations, real-space densities, local spectroscopy, conductivity and carrier diffusion within a unified stochastic framework, and computationally accessible.

We establish the accuracy and computational performance of RSATB through systematic comparisons with generalized diagonalization for insulating, semiconducting and metallic materials. The method reproduces density of states, charge density, local spectra and electronic response functions, with relative errors below 1% for the total density of states and below 10% for all benchmarked observables. Its computational work



Preconditioner	1,024 atoms		2,048 atoms	
	iters	#Ops	iters	#Ops
None	629	630.7	776	775.9
Jacobi	629	632.6	776	779.1
Block Jacobi	628	631.7	776	780.6
IC(0)	556	1,699.4	651	1,979.2
GAMG	240	1,221.8	261	1,327.7
BoomerAMG	344	5,853.9	329	6,105.6
ParaSails	311	334.0	485	526.6
Exact Cholesky	*1*	*10.7*	—	—
$Z = S^{-1/2}$ **(ours)**	**3**	**18.2**	**4**	**28.9**

Figure 2: **Different overlap inverse application method and the velocity operator treatment. a**, Conjugate-gradient iterations needed to apply S^{-1} preconditioned by our factorized Z on three typical basis set of the diamond 1,024 atoms system. **b** Conjugate-gradient iterations and number of equivalent matrix–vector products needed to apply S^{-1} in diamond with 1,024 and 2,048 atoms, for standard preconditioners and the $Z \approx S^{-1/2}$ preconditioner of RSATB (blue shaded). **c–d** The optical conductivity computed on the diamond system with our velocity implementation and two compared conventional approximation.

exhibits near-linear scaling, with a measured exponent of 1.01, up to 1,004,760 atoms and 13.1 million orbitals. To demonstrate the scientific opportunities enabled by this capability, we apply RSATB to a reconstructed alternating-twist trilayer graphene supermoiré containing 455,140 atoms and 5.92 million orbitals, using a machine-learned non-orthogonal Hamiltonian. The calculation resolves spatially heterogeneous flat-band spectra over an entire supermoiré period and reveals an energy-dependent redistribution of spectral weight between reconstructed plaquette and twiston regions, allowing direct comparison with spatially resolved tunnelling spectroscopy.

RSATB establishes a computational connection between large-scale learned electronic Hamiltonians and experimentally interpretable observables. By removing the diagonalization and overlap-metric bottlenecks, it enables electronic-structure information derived from first-principles calculations to be propagated to realistic atomistic environments without replacing their structural complexity by idealized or coarse-grained models. Beyond the materials demonstrated here, this capability provides a route to investigating how disorder, interfaces, reconstruction and nanoscale heterogeneity shape electronic properties, and to extracting microscopic information for larger-scale materials and device models.

Results

Scalable observable evaluation in a non-orthogonal basis requires a representation that does not depend on individual eigenstates. For the single-particle Schrödinger equation $HC = SC\Lambda$, the overlap matrix $S_{ij} = \langle\phi_i|\phi_j\rangle$ defines the metric of the localized orbitals. Introducing the dual basis $\{|\phi^i\rangle\}$, with $\langle\phi^i|\phi_j\rangle = \delta_{ij}$, gives the dual Hamiltonian matrix $\overline{H} = S^{-1}H$. Matrix functions of $\overline{H}$ then encode the spectrum, occupations and response information of the whole system.

RSATB evaluates these matrix functions with random-state estimators and polynomial expansions, reducing the calculation to repeated actions of the Hamiltonian and overlap operators on a small block of vectors. Both matrices are block sparse in a localized-orbital basis, so their application cost grows linearly with system size. The same recurrences yield global spectra, spatial distributions and time-dependent correlation functions. Whether this formal scaling is useful in practice depends on two non-orthogonal operations: repeated application of S^{-1} when the overlap is ill-conditioned, and construction of the position and velocity operators required for optical and transport response, which are addressed both below.

A reusable graph-local inverse factor accelerates stochastic propagation

The main practical obstacle to scalable calculation in a non-orthogonal random-state calculation is the inverse-overlap action $y = S^{-1}Hx$ at every polynomial step. A direct implementation first forms $u = Hx$ and then solves $Sy = u$. For an accurate LCAO basis, unavoidable near-linear dependencies lead to huge condition number and broad overlap spectrum. For diamond in the double-ζ polarized numerical atomic-orbital basis used here, we measure $\kappa(S) \simeq 8.7 \times 10^3$, which leads to the conjugate-gradient error

$$\frac{\|e_k\|_S}{\|e_0\|_S} \leq 2\left(\frac{\sqrt{\kappa(S)}-1}{\sqrt{\kappa(S)}+1}\right)^k, \tag{1}$$

where e_k is the error after k iterations and $\kappa(S)$ is the spectral condition number. With large $\kappa(S)$, e_k decay very slow with CG iterations. At the measured condition number, one overlap inverse action can entail hundreds of sparse overlap applications. Multiplied with the Chebyshev recurrence that may require thousands of these actions, this large prefactor can dominate the observable calculation even though each sparse operation scales linearly.

RSATB replaces these repeated solves with a sparse factor $Z \approx S^{-1/2}$ constructed once per overlap matrix. The elements of the inverse square root decay rapidly with graph distance in a localized basis, allowing a controlled sparse approximation. We initialize the factor from subgraph inverse-square-root on atomic neighbourhoods. The initial Z_0 can be used directly as preconditioner, and can be further improved by refining it with a specialized purification loop. The resulting factor Z is reused for every random state, polynomial step and observable.

Fig. 2a demonstrates the effectiveness of our method across three basis set of diamond, whose overlap's conditional number range from moderate to extreme $(28.7 - 227,000)$. Our factorized $Z \approx S^{-1/2}$ solve the metric application within 4 iteration for all. Fig. 2b compares our method with widely used alternatives on two overlap matrices in diamond systems of different sizes. Here, *iters* denotes the number of CG iterations required for convergence, while *#Ops* measures computational cost in units of overlap-matrix–vector multiplications. The unpreconditioned systems require 629 and 776 iterations, whereas the localized factor converges in three and four iterations. Conventional diagonal and incomplete-factorization preconditioners provide only modest improvements or introduce substantially more operator work. Although constructing the inverse factor entails an initial cost, its reuse across many Chebyshev steps and multiple observables substantially reduces the cumulative cost of metric operations, making repeated inverse-overlap applications inexpensive.

Sparse non-orthogonal velocity operators enable electronic response

Optical conductivity, electronic transport and carrier diffusion require a physically consistent description of charge dynamics. At large scales, calculations of these response functions can include the disorder, reconstruction and structural heterogeneity present in experiments. Their evaluation within Kubo linear-response theory requires the physical velocity operator

$$\hat{v}_\alpha = \frac{1}{\hbar}\hat{\mathcal{V}}_\alpha, \qquad \hat{\mathcal{V}}_\alpha \equiv i[\hat{H}, \hat{X}_\alpha]. \tag{2}$$

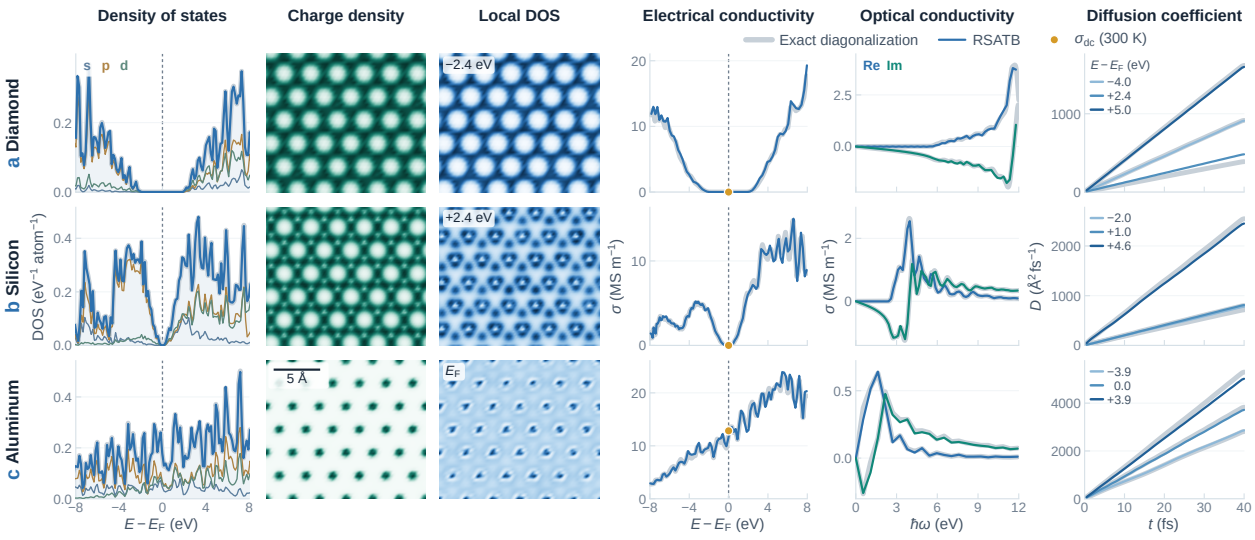
In a non-orthogonal atomic-orbital basis, the matrix representation of this commutator involves both the overlap metric and intra-orbital position contributions. The non-periodicity of the position operator further complicates its direct evaluation. We implement a sparse, periodic, metric-consistent representation that can be applied within the same random-state framework, enabling scalable evaluation of electronic response properties.

In reciprocal space, the matrix form of the velocity $\hat{\mathcal{V}}_\alpha = \hbar\hat{v}_\alpha$ is

$$\mathcal{V}_\alpha(\boldsymbol{k}) = i\Big(\mathcal{M}_\alpha[H] - \mathcal{M}_\alpha[S]\, S^{-1}H - S\big[S^{-1}\Delta_\alpha, S^{-1}H\big]\Big), \tag{3}$$

$$\mathcal{M}_\alpha[O](\boldsymbol{k})_{ij} \equiv \sum_{\boldsymbol{R}} d_{ij,\alpha}(\boldsymbol{R}) O_{ij}(\boldsymbol{R}) e^{-i\boldsymbol{k}\cdot\boldsymbol{d}_{ij}(\boldsymbol{R})} = i\,\partial_{k_\alpha} O(\boldsymbol{k}), \tag{4}$$

$$\boldsymbol{d}_{ij}(\boldsymbol{R}) \equiv \boldsymbol{R} + \boldsymbol{r}_j - \boldsymbol{r}_i, \qquad X_{\alpha,ij}(\boldsymbol{R}) = \langle \phi_{i\boldsymbol{0}} | \hat{X}_\alpha | \phi_{j\boldsymbol{R}} \rangle = (R + r_j)_\alpha S_{ij}(\boldsymbol{R}) + \Delta_{\alpha,ij}(\boldsymbol{R}). \tag{5}$$



d

Relative error	DOS	PDOS	$n(\mathbf{r})$	LDOS	$\sigma(E)$	Re $\sigma(\omega)$	Im $\sigma(\omega)$	$D(E,t)$
Diamond	0.70%	1.0%	0.35%	<0.01%	5.5%	4.2%	5.3%	4.9%
Silicon	0.58%	1.3%	2.7%	6.9%	5.3%	3.1%	3.5%	2.7%
Aluminum	0.81%	2.2%	3.6%	6.5%	3.4%	9.0%	9.7%	5.8%

Figure 3: **Accurate observables across insulating, semiconducting and metallic systems. a–c**, Total and orbital-projected density of states, charge density, local density of states (LDOS), energy-resolved electrical conductivity, complex optical conductivity and diffusion coefficient for diamond, silicon and aluminium, respectively. Blue and green curves show RSATB; grey curves show generalized-diagonalization references evaluated with matched temperature, broadening and energy resolution. Orange markers indicate the 300-K dc conductivity. **d**, Relative errors of the corresponding RSATB observables.

Here $\Delta_\alpha(\boldsymbol{R})$ is the bounded dipole contribution to the position matrix, evaluated from two-centre integrals (with two-centre library Op2c developed for RSATB). The physical velocity matrix used in the response functions is $v_\alpha(\boldsymbol{k}) = \mathcal{V}_\alpha(\boldsymbol{k})/\hbar$. In practice, we reorder the terms to further reduce compute within a few matrix-vector-product, details are introduced in method section.

Here we compare some commonly used formula for velocity operator in the Optical conductivity calculation, and compare the result with our exact formula. As shown in Fig.2c, two approximated method (Peierls [30]and Mulliken [31]) perform very differently. The Peierls method use directly the k-space derivative form [30], it does not capture the physics. The other approximation adopt the metric correction but use an approximated intra-orbital position [31]. It aligns with our result better at low energy spectrum of the real part, but derivative a lot in others. This calculation shows the significant importance to adopt the full component of the velocity operator to include metric correction and the full intra-orbital position.

Unified accuracy across electronic observables

We benchmark the random-state formulation against generalized diagonalization for an insulator (diamond), a semiconductor (silicon) and a metal (aluminium). In each comparison, RSATB and the reference calculation use the same H, S, temperature, spectral broadening and energy grid. With identical electronic operators and matched physical parameters, these comparisons quantify the numerical accuracy of the RSATB observable-evaluation framework.

The DOS is written as the stochastic trace

$$D(E) = g_s \,\mathrm{Tr}\big[\delta_\eta(E-\overline{H})\big] \approx \frac{g_s}{N_s}\sum_{s=1}^{N_s}\sum_{m=0}^{M-1} c_m(E,\eta)\langle\xi_s|T_m(\widetilde{H})|\xi_s\rangle, \tag{6}$$

where g_s is the spin degeneracy, δ_η is a broadened delta function, N_s is the number of samples and M is

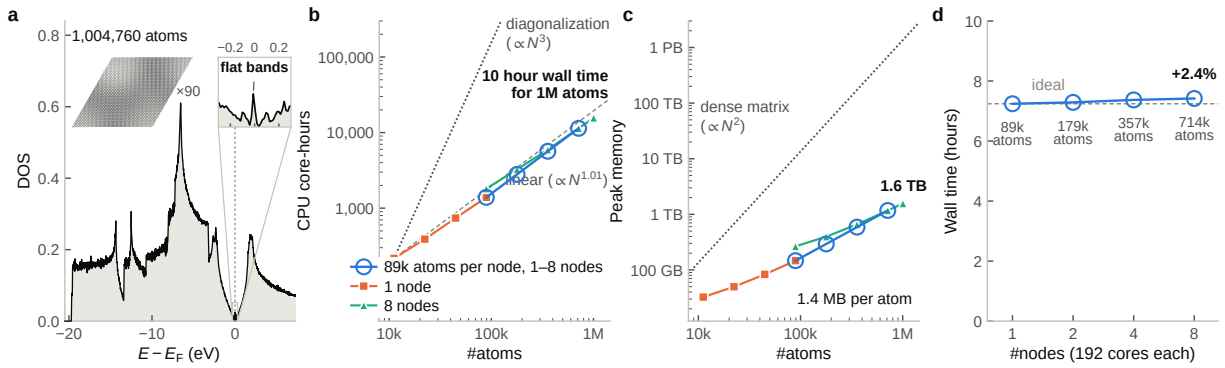


Figure 4: **Linear scaling of RSATB for a million-atom material. a**, Density of states of 1.08° twisted bilayer graphene with 1,004,760 atoms computed by RSATB; insets show the moiré stacking and the flat-band peak. **b–d**, Computing cost, peak memory and time for DOS runs on one node (orange squares), eight nodes (green triangles) and 89,312 atoms per node on 1–8 nodes (blue circles; fitted exponent 1.01). Grey dotted lines show conventional diagonalization reference; grey dashed lines show ideal linear scaling and, in **d**, the memory of eight nodes.

the polynomial order. The standard random coefficient vectors satisfy $\mathbb{E}[|\xi_s\rangle\langle\xi_s|] = I$; T_m and c_m are the Chebyshev polynomials and expansion coefficients, respectively, and $\widetilde{H}$ is the rescaled dual Hamiltonian defined in Methods. Increasing N_s reduces the statistical error, while increasing M improves the spectral resolution. Here, the eigenstates of H never appear, and only applications of $\overline{H} = S^{-1}H$ to vectors are needed [32, 33]. Projected spectra and response functions follow from the same principle by inserting the relevant projection or response operators into the stochastic trace.

Real-space observables also require random coefficient vectors that respect the overlap metric. Writing $\Phi(\mathbf{r})$ for the row vector of localized orbitals evaluated at $\mathbf{r}$, the electron density and LDOS are estimated as

$$n(\mathbf{r}) = g_s \mathbb{E}\left[\left|\Phi(\mathbf{r}) f_{\mathrm{FD}}^{1/2}(\overline{H}; \mu, T)|\zeta_s\rangle\right|^2\right], \tag{7}$$

$$\rho(\mathbf{r}, E) = g_s \mathbb{E}\left[\left|\Phi(\mathbf{r}) \delta_\eta^{1/2}(E - \overline{H})|\zeta_s\rangle\right|^2\right], \qquad \mathbb{E}[|\zeta_s\rangle\langle\zeta_s|] = S^{-1}. \tag{8}$$

Here μ denotes the chemical potential and $|\zeta_s\rangle$ denotes a metric-aware random coefficient vector, sampled using the overlap factor Z introduced below. We use $N_s = 64$ for the benchmark calculations. As shown in Fig. 3a–c, RSATB reproduces the total and orbital-projected DOS, the real-space charge and LDOS patterns across all three materials. Global traces benefit most from self-averaging: the DOS relative errors are 0.58–0.81% and the projected-DOS relative errors are 1.0–2.2%. Charge-density relative errors are 0.35–3.6%, and local DOS relative errors remain below 7%. These results demonstrate quantitative agreement across spectral and real-space observables, with stochastic errors systematically reducible by increasing the number of random states.

The remaining columns of Fig. 3 show the energy-resolved electrical conductivity, complex optical conductivity and diffusion coefficient for the same three systems, enabled by the velocity formalism. The electrical and optical conductivities use Kubo formulations, accelerated where appropriate by Fourier–Chebyshev contraction [15]; the diffusion coefficient is extracted from the mean-square displacement. Relative errors range from 2.7% to 9.7%, showing that the accuracy of the non-orthogonal formulation extends from the DOS to transport observables.

Near-linear scaling to one million atoms

Besides the overlap metric application, the remaining cost bottleneck is mainly caused by the dense neighbour graph of the LCAO matrices. Within typical cutoffs of 7–10 Å, an atom can couple to hundreds of neighbours, producing dense atomic blocks within a globally sparse structure that is costly to describe by conventional

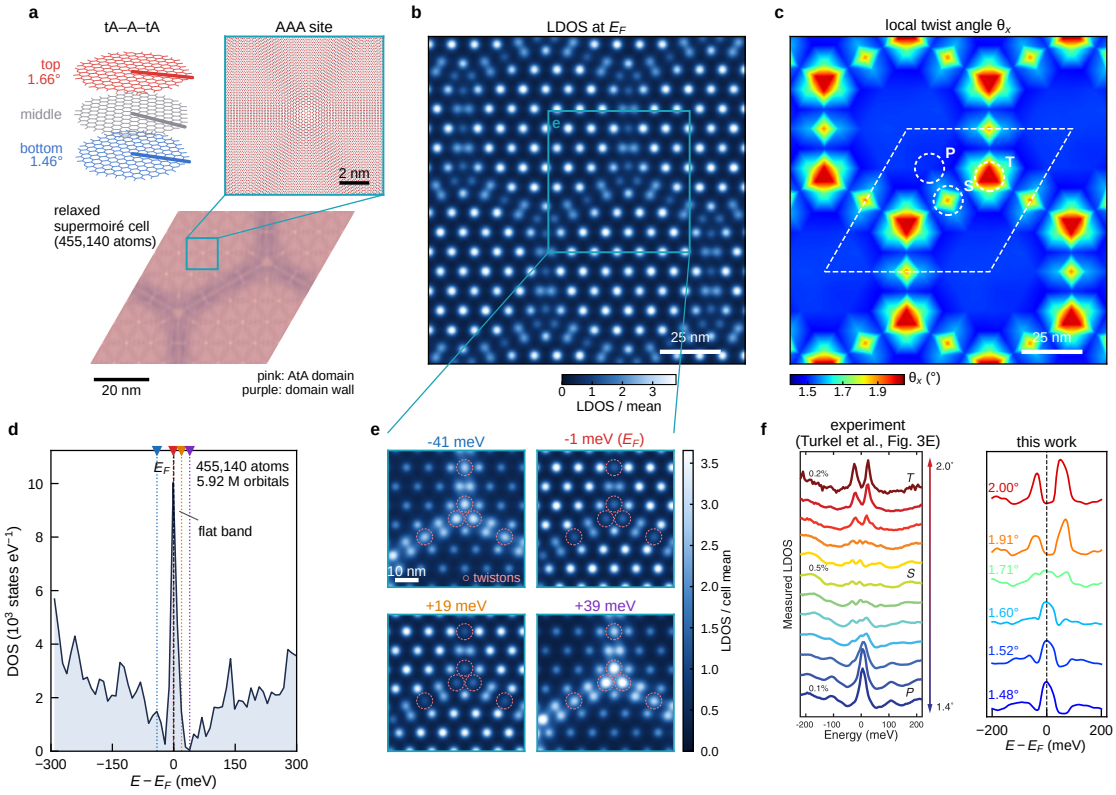


Figure 5: **Real-space flat-band spectroscopy of a 455,140-atom twisted-trilayer-graphene supermoiré.** **a**, Relaxed 67.8-nm tA–A–tA supercell and atomic structure around an AAA site. **b**, LDOS at charge neutrality, spatially averaged with a 1.2-nm Gaussian tip. **c**, Local twist angle θ_x, obtained from the Voronoi area of the AAA sites following ref. [35]; P, S and T denote plaquette, soliton and twiston regions. **d**, Total DOS of the supermoiré system; triangles mark the energies shown in **e**. **e**, Energy-dependent LDOS around a twiston cluster. Plaquettes dominate near E_F, whereas twistons gain spectral weight away from charge neutrality. **f**, Experimental AAA-site spectra from ref. [35] and calculated spectra grouped by the local twist angle θ_x.

sparse matrix. Therefore, we develop VBCSR as the backend for RSATB, which is a distributed variable-block sparse matrix library dedicated for the sparse linear algebra action of such a sparse global atomic operator with locally variable dense block structure [34]. This layout avoids scalar sparse-matrix inefficiency for LCAO-like operator, distributes storage and computation, and exposes block operations to highly optimized kernels.

We test whether the local formulation retains its computational advantage as the system grows by calculating the DOS of twisted bilayer graphene at $\theta \simeq 1.08°$. The moiré cell contains 11,164 atoms, and its in-plane supercells from 1×1 to 10×9 have 11,164 to 1,004,760 atoms, corresponding to 1.45×10^5 to 1.31×10^7 orbitals in the C 2s2p1d basis. DeePTB predicts the non-orthogonal ab-initio LCAO Hamiltonian for each supercell. Every calculation uses the same 31,503 Chebyshev moments and four random vectors at the Γ point. Under these fixed spectral-resolution and stochastic-sampling parameters, RSATB exhibits near-linear computational work with system size.

The computational work scales nearly linearly with system size (Fig.4b), while peak memory exhibits a similar trend (Fig.4c). Weak-scaling tests at a fixed workload of 89,312 atoms per node show wall-time variations within 2.5% across one to eight nodes (Fig.4d), demonstrating efficient parallel scalability.

Flat-band spectral weight redistributes across the supermoiré

Alternating-twist trilayer graphene develops a hierarchy of structural length scales when the two twist angles differ slightly. Each interface forms a moiré pattern on the nanometre scale, while the mismatch between the two patterns produces a much longer supermoiré modulation (see Fig. 5a). Scanning tunnelling microscopy (STM) shows that structural relaxation reorganizes this modulation into extended domains separated by solitons and localized twist-angle defects, or twistons, together with pronounced spatial variations in the low-energy spectrum [35]. For experimentally relevant twist angles, a single supermoiré period approaches 70 nm and contains hundreds of thousands of atoms. Resolving its electronic structure therefore requires an atomistic calculation far beyond the range of conventional generalized diagonalization.

We calculate a complete reconstructed tA–A–tA supermoiré cell with twist angles of 1.456° and 1.664°. Their 0.208° mismatch produces a 67.8-nm periodic cell containing 455,140 atoms after relaxation with a machine-learned interatomic potential (Fig. 5a). The reconstruction forms distinct plaquettes, solitons and twistons, resolved by the local twist field θ_x obtained from neighbouring AAA sites (Fig. 5c). We then predict the full non-orthogonal LCAO Hamiltonian with DeePTB [2, 4]. With 13 orbitals per atom, the resulting operator contains 5.92×10^6 orbitals. RSATB evaluates its total and local spectra directly.

The reconstructed structure produces a strongly inhomogeneous low-energy electronic state. At charge neutrality, the LDOS varies across the complete supermoiré cell and reveals the structural reconstruction with charge distribution (Fig. 5b). The total DOS contains a narrow flat-band feature around E_F (Fig. 5d), but the associated spectral weight is not spatially uniform. Over an energy window of only a few tens of millielectronvolts, it moves between different parts of the reconstruction: the relaxed plaquettes are brightest near E_F, whereas the twistons gain spectral weight away from charge neutrality (Fig. 5e). A single global flat-band peak therefore contains distinct local electronic environments that become visible only in the real-space calculation.

We next compare the atomistic spectra with local scanning tunnelling microscopy measurements. For each AAA site, we extract the LDOS and assign a local twist angle θ_x using the geometric measure applied to the experimental structure [35]. Grouping the calculated spectra by θ_x reveals a systematic evolution across the experimentally sampled twist-angle range (Fig. 5f). The calculation reproduces the observed progression from a sharp plaquette spectrum to broadened and split low-energy features through the soliton and twiston environments.

This calculation brings learned electronic-structure models to the spatial scale of the experiment. Starting from the reconstructed atomic structure, the learned Hamiltonian and RSATB yield the total spectrum, spatially resolved LDOS and site-resolved spectroscopy for a 5.92-million-orbital system. This links local atomic structure directly to the measured electronic response at modest cost, positioning this work as a bridge that makes experiment-scale properties accessible from first-principles.

Discussion

RSATB provides a scalable framework for extracting electronic observables directly from large, non-orthogonal localized-orbital Hamiltonians. Its central advance rests on overcoming two complementary computational bottlenecks: the repeated application of the inverse overlap metric and the scalable evaluation of physically consistent velocity operators. A reusable graph-local inverse factor makes stochastic propagation efficient, while the sparse, metric-consistent velocity implementation extends the same framework to optical and electronic transport response. Benchmark calculations establish agreement with generalized diagonalization, and near-linear computational work up to 13.1 million orbitals demonstrates the practical scalability of the approach. The reconstructed twisted-trilayer-graphene application further shows how learned Hamiltonians and RSATB can connect atomistic structural complexity with spatially resolved electronic spectroscopy at experimentally relevant scales.

Connecting microscopic theory with experiment. The immediate consequence is access to electronic observables in structures that were previously too large to treat without severe coarse-graining or idealization. Disorder, interfaces, extended defects, reconstructed domains and other mesoscale structural motifs can therefore be retained explicitly while computing quantities such as DOS, LDOS, charge density and linear-response functions. This enables microscopic calculations to be compared with spatially resolved experiments

on the same structural landscape, providing interpretation of measured signals, and mechanistic understanding by access to layer, orbital and real-space information that experiments cannot readily resolve directly. As the accuracy of the underlying Hamiltonian improves, the same framework can transfer those advances directly to experimentally relevant length scales.

Bridging microscopic and engineering models. The same capability opens a route from electronic structure to reduced models used at larger scales. Device and continuum simulations necessarily describe microscopic physics through effective quantities such as carrier densities, dielectric response, conductivity, capacitance or other constitutive parameters, which are commonly extracted from idealized calculations and subsequently calibrated against experiment. RSATB makes it possible to evaluate these quantities in structurally realistic and heterogeneous environments. Such microscopic information could help constrain reduced models and improve predictive materials and device simulations.

More broadly, RSATB closes a gap between local electronic-structure learning and large-scale physical prediction. The resulting hierarchy, from first-principles data, to learned Hamiltonians, to observable evaluation, and ultimately to reduced models and experiment, provides a route for microscopic electronic structure to participate directly in the interpretation and design of imperfect and realistic materials and devices. We expect this capability to expand the range of questions for which atomistic electronic theory can be applied, from explaining emergent phenomena in complex quantum materials to informing multiscale models for materials and device engineering.

Methods

Random-state evaluation of observables

Polynomial and random-state engine

We use the following notation throughout. A hat denotes an abstract Hilbert-space operator, whereas the corresponding unhatted symbol denotes its covariant matrix in the LCAO basis; an overbar marks the dual matrix representation, and a tilde marks spectral rescaling. Thus, H and S are the covariant Hamiltonian and overlap matrices, $\overline{H} = S^{-1}H$ is the dual Hamiltonian matrix and $\widetilde{H}$ is its rescaling to $[-1, 1]$. Standard random coefficient vectors are denoted by $|\xi_s\rangle$ and satisfy $\mathbb{E}[|\xi_s\rangle\langle\xi_s|] = I$; metric-aware random coefficient vectors are denoted by $|\zeta_s\rangle$ and satisfy $\mathbb{E}[|\zeta_s\rangle\langle\zeta_s|] = S^{-1}$. The index $s = 1, \ldots, N_s$ labels stochastic samples, M is the Chebyshev order, N_{orb} is the orbital dimension, μ is the chemical potential and η is the spectral broadening.

The random-state method avoids diagonalization by reducing matrix functions and traces to operator actions on a small set of vectors. In a non-orthogonal LCAO basis, the overlap metric is incorporated through:

$$\overline{H} = S^{-1}H, \qquad \widetilde{H} = \frac{\overline{H} - E_{\mathrm{c}} I}{\Delta E}, \qquad \Delta E = \frac{E_{\max} - E_{\min}}{2}, \quad E_{\mathrm{c}} = \frac{E_{\max} + E_{\min}}{2}, \tag{9}$$

where $E_{\min}$ and $E_{\max}$ bound the generalized spectrum of (H, S), such that the spectrum of $\widetilde{H}$ lies strictly inside $[-1, 1]$. A function f of the Hamiltonian is approximated by the finite Chebyshev series

$$f(\overline{H})|\xi_s\rangle \simeq \sum_{m=0}^{M-1} c_m[f]\,|t_{m,s}\rangle, \qquad |t_{0,s}\rangle = |\xi_s\rangle, \quad |t_{1,s}\rangle = \widetilde{H}|\xi_s\rangle, \quad |t_{m+1,s}\rangle = 2\widetilde{H}|t_{m,s}\rangle - |t_{m-1,s}\rangle. \tag{10}$$

The coefficients $c_m[f]$ are obtained by a cosine transform of the rescaled scalar function $x \mapsto f(\Delta E\, x + E_{\mathrm{c}})$ [16]. For a localized sparse operator, each recurrence step is a sparse matrix–vector or matrix–block product with $O(N_{\mathrm{orb}})$ cost at fixed sparsity.

Stochastic trace estimation gives

$$\mathrm{Tr}\, B = \mathbb{E}\langle\xi_s|B|\xi_s\rangle \simeq \frac{1}{N_s}\sum_{s=1}^{N_s}\langle\xi_s|B|\xi_s\rangle. \tag{11}$$

For independent random vectors with zero mean and unit covariance, this estimator is unbiased and its standard error decreases as $N_s^{-1/2}$ [32, 33]. Extensive traces also self-average the standard error as $N_{\mathrm{orb}}^{-1/2}$, whereas local quantities generally retain a larger sampling variance [16, 17, 19].

Spectra, occupations and chemical potential

For the density of states (DOS), we first accumulate the moments $\gamma_m = N_s^{-1}\sum_s \langle\xi_s|T_m(\widetilde{H})|\xi_s\rangle$ and then reconstruct

$$D_M(E) = \frac{1}{\pi\Delta E\sqrt{1-\epsilon^2}}\left[J_0\gamma_0 + 2\sum_{m=1}^{M-1} J_m\gamma_m T_m(\epsilon)\right] g_s, \qquad \epsilon = \frac{E-E_{\mathrm{c}}}{\Delta E}, \tag{12}$$

where J_m are Jackson damping factors that suppress truncation oscillations.

Projected DOS (PDOS) is obtained during the same recurrence using

$$\gamma_m^{(\mathcal{G})} = \mathbb{E}\langle\xi_s|\Pi_{\mathcal{G}} T_m(\widetilde{H})|\xi_s\rangle, \tag{13}$$

where the dual-basis mask $\Pi_{\mathcal{G}}$ selects an atom, element or orbital group $\mathcal{G}$.

Carrier filling and band energy use the same moment recursion with coefficients obtained from the Fermi–Dirac function,

$$N_e = g_s \,\mathrm{Tr}\, f_{\mathrm{FD}}(\overline{H};\mu,T), \qquad E_{\mathrm{band}} = g_s \,\mathrm{Tr}\big[\overline{H} f_{\mathrm{FD}}(\overline{H};\mu,T)\big]\,. \tag{14}$$

Once the moments have been accumulated, both quantities can be evaluated for a new chemical potential without additional Hamiltonian actions. The chemical potential is therefore obtained efficiently by bisection.

Real-space properties and metric-aware sampling

A global trace self-averages over the full system, whereas a real-space observable is estimated independently at each spatial point. For any non-negative spectral function g, define

$$\mathcal{P}_g = g(\overline{H})S^{-1}, \qquad \mathcal{F}_g = g^{1/2}(\overline{H}), \qquad q_g(\boldsymbol{r}) = g_s\,\mathbb{E}\,|\Phi(\boldsymbol{r})\mathcal{F}_g|\zeta_s\rangle|^2 = g_s\,\Phi(\boldsymbol{r})\mathcal{P}_g\Phi(\boldsymbol{r})^\dagger. \tag{15}$$

Here $\Phi(\boldsymbol{r})$ contains the values of the localized orbitals at the requested real-space points. Taking $g(x) = \delta_\eta(E-x)$ gives the broadened local density of states $\rho(\boldsymbol{r},E)$, and taking $g(x) = f_{\mathrm{FD}}(x;\mu,T)$ gives the electron density $n(\boldsymbol{r})$. All requested energies or occupation windows share one Chebyshev recurrence; only the scalar expansion coefficients change.

The metric-aware vector $|\zeta_s\rangle$ is generated from the standard vector $|\xi_s\rangle$ using the sparse factor Z, for which $Z^\dagger SZ \approx I$:

$$|\zeta_s\rangle = Zh(W)|\xi_s\rangle, \qquad W = Z^\dagger SZ, \qquad h(W) \simeq W^{-1/2}. \tag{16}$$

For the exact correction $h(W) = W^{-1/2}$, its covariance is

$$\mathbb{E}[|\zeta_s\rangle\langle\zeta_s|] = ZW^{-1/2}\mathbb{E}[|\xi_s\rangle\langle\xi_s|]W^{-1/2}Z^\dagger = S^{-1}. \tag{17}$$

In practice, $h(W)$ is converged to the requested sampling tolerance. Because $W \approx I$, the polynomial or Krylov correction requires only a low polynomial order with our approximated factorization Z used for preconditioning.

Optical and electronic response

Linear-response functions describe the electronic current induced by an external perturbation. For independent electrons, the optical conductivity is evaluated from the time-domain current correlation [8]

$$\begin{aligned}\sigma_{\alpha\beta}(\omega) &= \frac{2g_s e^2}{\hbar\Omega}\frac{e^{-\beta_T\hbar\omega}-1}{\omega}\int_0^\infty \mathrm{d}t\; w(t)\sin(\omega t)\;\mathrm{Im}\,\mathrm{Tr}\Big[f_{\mathrm{FD}}(\hat{H};\mu,T)\hat{v}_\beta(t)[\hat{I}-f_{\mathrm{FD}}(\hat{H};\mu,T)]\hat{v}_\alpha\Big]\,,\\ \hat{v}_\beta(t) &= \hat{U}(t)^\dagger\hat{v}_\beta\hat{U}(t), \qquad \hat{U}(t) = e^{-i\hat{H}t/\hbar}, \qquad \hat{v}_\alpha = \frac{i}{\hbar}[\hat{H},\hat{X}_\alpha].\end{aligned} \tag{18}$$

Here Ω is the system volume, $\beta_T = (k_{\mathrm{B}}T)^{-1}$ and $w(t)$ is a finite-time damping window; we use a smooth Gaussian window. The time propagation is accelerated with the concurrent stochastic-propagation method of ref. [36].

In the dual representation, non-orthogonality modifies the matrix representations of the Hamiltonian, velocity and time evolution in Eq. (18):

$$\overline{v}_\alpha = S^{-1}v_\alpha, \qquad \overline{v}_\beta(t) = \overline{U}(t)^{-1}\overline{v}_\beta\overline{U}(t), \qquad \overline{U}(t) = e^{-i\overline{H}t/\hbar}, \qquad \overline{U}(t)^{-1} \quad = S^{-1}\overline{U}(t)^\dagger S. \tag{19}$$

For electronic transport, the longitudinal spectrum is defined by the Kubo–Greenwood expression

$$\sigma^{\mathrm{KG}}_{\alpha\alpha}(E) = \frac{\pi g_s e^2\hbar}{\Omega}\,\mathrm{Tr}\Big[\hat{v}_\alpha\delta_\eta(E-\hat{H})\hat{v}_\alpha\delta_\eta(E-\hat{H})\Big]. \tag{20}$$

Here η is the spectral broadening. For the full conductivity tensor, we evaluate the Kubo–Bastin form [37–39]:

$$\sigma^{\mathrm{KB}}_{\alpha\beta}(\mu,T) = -\frac{2g_s e^2\hbar}{\Omega}\int \mathrm{d}E\, f_{\mathrm{FD}}(E;\mu,T)\mathrm{Im}\,\mathrm{Tr}\left[\hat{v}_\alpha\frac{\partial G_\eta(E)}{\partial E}\hat{v}_\beta\delta_\eta(E-\hat{H})\right], \qquad G_\eta(E) = (E-\hat{H}+i\eta)^{-1}. \tag{21}$$

In the non-orthogonal dual representation, $\hat{H}$, $\hat{v}$ and G in Eqs. (20)–(21) are represented by $\overline{H}$, $\overline{v} = S^{-1}v$ and $\overline{G} = (E-\overline{H}+i\eta)^{-1}$, respectively.

Forming the usual two-index table of M Chebyshev moments would require $O(N_{\mathrm{orb}}M^2)$ work and an explicit $M\times M$ tensor. We instead use the Fourier–Chebyshev contraction [15] to reformulate the calculation as Fourier transforms and contractions of a vector sequence, reducing the cost to $O(N_{\mathrm{orb}}M\log M)$ and avoiding the double-moment table.

The mean-square displacement (MSD) measures the spatial spreading of carriers at energy E over time t:

$$\mathrm{MSD}_\alpha(E,t) = \frac{\mathrm{Tr}\Big[\hat{\mathcal{X}}_\alpha(t)^\dagger\delta_\eta(E-\hat{H})\hat{\mathcal{X}}_\alpha(t)\Big]}{\mathrm{Tr}[\delta_\eta(E-\hat{H})]}, \tag{22}$$

$$\hat{\mathcal{X}}_\alpha(t) = [\hat{X}_\alpha, \hat{U}(t)]. \tag{23}$$

Both traces are evaluated with the stochastic method with standard random state $|\xi_s\rangle$, under dual matrix representation. The time dependence of the MSD yields the propagation velocity, mean free path, diffusion coefficient and associated Kubo conductivity. We extract the diffusion coefficient from $D(E,t) = 0.5\times \mathrm{dMSD(E,t)/dt}$.

Sparsity-aware inverse-overlap factorization

We construct a sparse inverse factor of the non-orthogonal overlap by combining a graph-local matrix-function approximation with a sparsity-aware Newton–Schulz refinement.

Adaptive graph-based initialization

The construction begins from the sparsity graph $G(S)$. Let $\mathcal{N}_i$ denote the neighbourhood of atomic block i within graph radius R_{loc}. For each atom, we extract the dense principal submatrix supported on that neighbourhood, take its inverse square root, and keep only the columns belonging to the central atom:

$$A_i = S[\mathcal{N}_i,\mathcal{N}_i], \qquad Z_0[\mathcal{N}_i,i] = A_i^{-1/2}E_i, \tag{24}$$

where E_i selects the orbitals of atom i within $\mathcal{N}_i$. Assembling these columns gives a global sparse Z_0. Exponential decay bounds the truncation, $\left|[S^{-1/2}]_{ij}\right| \le Ce^{-\alpha d(i,j)}$, and the error is systematically controllable by increasing the finite radius R_{loc}.

Direct diagonalization of every neighbourhood remains expensive because $n_i^{\mathrm{loc}} = \dim\mathcal{N}_i$ can reach 10^3–10^4 orbitals. Equation (24), however, requires only the block action $A_i^{-1/2}E_i$ instead of the complete inverse

square root. We therefore apply block Lanczos to A_i^{-1} using two triangular solves with the factorization $A_i = LL^\dagger$. The m-step Krylov approximation is

$$A_i^{-1/2} E_i \;\approx\; Q_m \, K_m^{1/2} \, Q_m^\dagger E_i, \qquad K_m = Q_m^\dagger A_i^{-1} Q_m, \tag{25}$$

with K_m block tridiagonal of dimension $m n_i^{\mathrm{orb}} \ll n_i^{\mathrm{loc}}$, where n_i^{orb} is the number of orbitals on atom i.

Applying the Krylov method to A_i^{-1} exploits the near-linear dependence of the LCAO basis, and maps the small eigenvalues that dominate the inverse square root to the extremal part of the spectrum, accelerating their capture [40, 41].

Across synthetic spectra at $\kappa = 10^8$ this route is 3.5–11.6× faster than the eigendecomposition it replaces and reduces the error from 1.3×10^{-8} to 4×10^{-9}. The accuracy is gained by avoiding the error amplification of a full eigendecomposition of an ill-conditioned matrix.

Sparsity-aware refinement with dual routes

Once Z_0 is assembled, it is refined towards an inverse factor using

$$Z_{k+1} = Z_k P_k, \qquad P_k = \tfrac{15}{8} I - \tfrac{5}{4} W_k + \tfrac{3}{8} W_k^2, \qquad W_k = Z_k^\dagger S Z_k = P_{k-1} W_{k-1} P_{k-1}, \tag{26}$$

The polynomial is a high-order recursive inverse-factorization update [12]. Products are first filtered by a block threshold to keep sparsity, and convergence is monitored through the applied residual $\|(Z^\dagger S Z - I)v\|/\|v\|$ on random probes v. This residual requires only three sparse products and never forms the explicit dense $Z^\dagger S Z$. The dominant cost is the update of W_k.

The update has two algebraically equivalent routes: recomputing $W_k = Z_k^\dagger S Z_k$, or propagating it as $W_k = P_{k-1} W_{k-1} P_{k-1}$. Their sparsity evolves in opposite directions. The inverse factor Z_k fills as longer-range terms of $S^{-1/2}$ are accumulated, whereas W_k and P_k contract towards the identity. In a 4,096-atom cell, the average number of block neighbours per row of Z increases from 293 to 2,255 and then saturates, while that of P decreases from 1,628 to 552 and finally to 1. The direct $Z_k^\dagger S Z_k$ route is cheaper early in the refinement, whereas the propagated route becomes cheaper close to convergence.

RSATB measures the cost of each route during the calculation and updates an empirical model $C(\cdot)$. At each refinement step, it selects

$$\mathrm{route}_k = \arg\min \big\{ \, C(Z_k^\dagger S Z_k), \; C(P_{k-1} W_{k-1} P_{k-1}) \, \big\}, \tag{27}$$

For a 1,024-atom diamond system, adaptive routing reduces the refinement time from 955.8 to 223.2 s, a factor of 4.3 relative to the fixed route.

Fused sparse kernels without dense intermediates

The formal intermediates in Eq. (26) can be substantially denser than the filtered result. Squaring W_k doubles its sparsity radius even as the polynomial P_k contracts towards the identity. Similarly, forming PW before PWP or forming SZ before $Z_k^\dagger S Z_k$ can determine the peak memory and arithmetic cost.

We avoid these intermediates with fused kernels for $ABA^\dagger$, $A^\dagger BA$ and the polynomial $c_2 A^2 + c_1 A + c_0 I$. Required blocks are streamed and filtered as they are generated, so the complete intermediate is never stored. Hermiticity of W_k and P_k halves their storage and reduces the associated arithmetic, while $Z_{k+1} = Z_k P_k$ is updated in place so that two complete copies of Z do not coexist. These kernels are implemented in the distributed variable-block sparse library VBCSR [34].

Graph-local initialization gives a good starting point, adaptive routing reduces refinement cost, and fused execution controls intermediate fill. These components make the inverse-overlap factor practical at the scales RSATB is targeting.

Velocity in the non-orthogonal orbital basis

To keep factors of $\hbar$ explicit, we derive the velocity numerator $\hat{\mathcal{V}}_\alpha \equiv \hbar \hat{v}_\alpha$:

$$\hat{\mathcal{V}}_\alpha = i[\hat{H}, \hat{X}_\alpha], \qquad \hat{v}_\alpha = \hat{\mathcal{V}}_\alpha / \hbar. \tag{28}$$

Let orbital i be centred at $\boldsymbol{r}_i$ and orbital j in cell $\boldsymbol{R}$ at $\boldsymbol{R}+\boldsymbol{r}_j$, and define their separation $\boldsymbol{d}_{ij}(\boldsymbol{R}) = \boldsymbol{R}+\boldsymbol{r}_j-\boldsymbol{r}_i$. In an orthogonal localized basis, the familiar result is the displacement-weighted Hamiltonian [26, 28],

$$\mathcal{M}_\alpha[O](\boldsymbol{k})_{ij} \equiv \sum_{\boldsymbol{R}} d_{ij,\alpha}(\boldsymbol{R}) O_{ij}(\boldsymbol{R}) e^{-i\boldsymbol{k}\cdot\boldsymbol{d}_{ij}(\boldsymbol{R})} = i\,\partial_{k_\alpha} O(\boldsymbol{k}),$$
$$\mathcal{V}_\alpha^{\mathrm{ortho}}(\boldsymbol{k}) = i\mathcal{M}_\alpha[H](\boldsymbol{k}) = -\partial_{k_\alpha} H(\boldsymbol{k}). \tag{29}$$

Thus the absolute position $\hat{X}_\alpha$ is avoided. This is essential for periodic calculations because the position operator is unbounded and cannot be applied directly to cell-periodic coefficient vectors.

This expression is incomplete when the orbitals overlap. Let H and S denote the Bloch-summed matrices at the same $\boldsymbol{k}$, and let X_α denote the corresponding formal position matrix. Resolving the identity in a non-orthogonal basis gives $\hat{I} = \sum_{ij} |i\rangle (S^{-1})_{ij} \langle j|$, so matrix products contain the inverse metric. Equation (28) is therefore represented exactly by

$$\mathcal{V}_\alpha = i\left(HS^{-1}X_\alpha - X_\alpha S^{-1}H\right). \tag{30}$$

For scalable application, we require an equivalent expression that avoids direct formation of the unbounded position matrix.

We separate each position matrix element about the centre of its ket orbital,

$$X_{\alpha,ij}(\boldsymbol{R}) = (R + r_j)_\alpha S_{ij}(\boldsymbol{R}) + \Delta_{\alpha,ij}(\boldsymbol{R}). \tag{31}$$

The first term locates the absolute position of the orbital; the bounded remainder Δ_α describes charge displacement within the extended orbitals. It is obtained from two-centre position integrals and cannot, in general, be reconstructed from H, S, and the atomic coordinates alone [26, 28]. Define its periodic form by

$$[X_\alpha(\boldsymbol{k})]_{ij} \equiv \sum_{\boldsymbol{R}} X_{\alpha,ij}(\boldsymbol{R}) e^{-i\boldsymbol{k}\cdot\boldsymbol{d}_{ij}(\boldsymbol{R})}, \qquad [\Delta_\alpha(\boldsymbol{k})]_{ij} \equiv \sum_{\boldsymbol{R}} \Delta_{\alpha,ij}(\boldsymbol{R}) e^{-i\boldsymbol{k}\cdot\boldsymbol{d}_{ij}(\boldsymbol{R})}. \tag{32}$$

In the following equations, all Bloch-summed matrices are evaluated at the same $\boldsymbol{k}$, which is omitted below for readability.

Substituting Eq. (32) and Eq. (31) into Eq. (30) and canceling the absolute orbital centres gives the periodic non-orthogonal representation

$$\mathcal{V}_\alpha = i\left\{\mathcal{M}_\alpha[H] - \mathcal{M}_\alpha[S]S^{-1}H - S\left[S^{-1}\Delta_\alpha, S^{-1}H\right]\right\}. \tag{33}$$

The first two terms account for the fact that both $H(\boldsymbol{k})$ and $S(\boldsymbol{k})$ change across reciprocal space. Indeed,

$$i\,\partial_{k_\alpha}(S^{-1}H) = S^{-1}\left\{\mathcal{M}_\alpha[H] - \mathcal{M}_\alpha[S]S^{-1}H\right\}. \tag{34}$$

The final commutator in Eq. (33) supplies the additional motion of charge within the orbitals. It vanishes from a band's diagonal velocity but remains in interband matrix elements and is therefore essential for optical response.

For sparse application, let $R^{\mathrm{at}}_{\alpha,ij} = r_{i,\alpha}\delta_{ij}$ and $X^{\mathrm{loc}}_\alpha = X_\alpha - R^{\mathrm{at}}_\alpha S$. Since $X^{\mathrm{loc}}_\alpha = \mathcal{M}_\alpha[S] + \Delta_\alpha$, we can express the action of the velocity on a coefficient vector c as

$$\mathcal{V}_\alpha c = i\left(\mathcal{M}_\alpha[H]c - X^{\mathrm{loc}}_\alpha S^{-1}Hc + HS^{-1}\Delta_\alpha c\right). \tag{35}$$

This is the form implemented in RSATB. Only the three bounded sparse matrices $\mathcal{M}_\alpha[H]$, X^{loc}_α, and Δ_α are stored; the two actions of S^{-1} use the same solve already available to the random-state calculation. We therefore never construct the unbounded position operator or a dense inverse overlap. In the orthogonal point-orbital limit, $S = I$ and $\Delta_\alpha = 0$, and Eq. (35) reduces to Eq. (29). For a normalized generalized eigenstate $Hc_n = \varepsilon_n S c_n$, it also gives $c_n^\dagger \mathcal{V}_\alpha c_n = -\partial_{k_\alpha}\varepsilon_n$, and hence $c_n^\dagger v_\alpha c_n = -(1/\hbar)\partial_{k_\alpha}\varepsilon_n$, providing an independent check of the metric placement, sign and units. This reorganization retains the exact non-orthogonal velocity while allowing its application with sparse matrices in large random-state calculations.

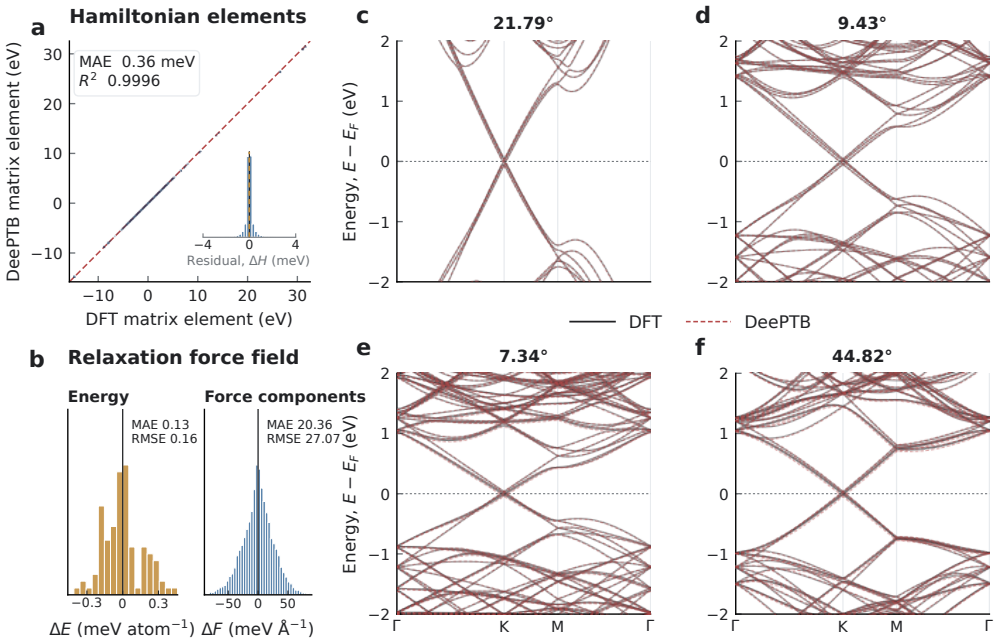


Figure 6: **Accuracy of the learned structural and electronic models.** **a**, DeePTB Hamiltonian matrix elements against DFT references; the inset shows the residual distribution. **b**, Energy and Cartesian force-component errors of the interatomic potential on held-out structures. **c**–**f**, Band structures of unseen twisted-trilayer-graphene structures at four twist angles. Solid black curves show DFT and dashed red curves show DeePTB predictions.

Machine-learned structural and electronic models

DeePTB Hamiltonian model

Direct self-consistent density-functional theory is not practical for the nearly 500,000-atom trilayer structure. We therefore use DeePTB to predict its sparse real-space Hamiltonian blocks directly in the same full, non-orthogonal LCAO basis used by RSATB [2, 4]. DeePTB represents the atoms and their neighbour relations as a graph and predicts the onsite and interatomic blocks $H_{ip,jq}(\boldsymbol{R})$, where i, j label atoms and p, q label their orbitals. The network is $E(3)$-equivariant, ensuring that the predicted blocks transform consistently with the atomic geometry and orbital symmetries.

We sample the structural configurations by active learning over the same configuration space used for the interatomic potential described below. Hamiltonian labels are computed from first-principles calculations with ABACUS [42]; `dftio` converts the atomic structures, basis information and matrix blocks into the format used for DeePTB training. We use the localized equivariant message-passing (LEM) architecture [4]. LEM updates many-body features through successive atomic neighbourhoods with a smooth, learned distance dependence, allowing a local Hamiltonian block to respond to stacking, strain and thermal distortion while retaining the required orbital symmetries. The training data contain bilayer and trilayer graphene structures spanning relative translations, temperatures and pressures. The validation data contain moiré structures at twist angles excluded from training, testing transfer from local environments to unseen moiré geometries. Figure 6a compares individual Hamiltonian elements with DFT and gives a mean absolute error of 0.36 meV. The band structures at four held-out twist angles (Fig. 6c–f) closely reproduce their DFT references over the displayed energy window.

To apply the trained model to the full structure, we extended DeePTB with distributed graph inference. Recursive inertial bisection divides the atoms into compact spatial domains. Each process stores the atoms it owns and only the neighbouring "ghost" atoms needed to complete interactions across its domain boundary. At each message-passing layer, processes exchange the current features of these boundary atoms; after the final layer, every Hamiltonian block pair (i, j) is returned to the process that owns its first atom. The parallel

and serial paths produce the same Hamiltonian up to floating-point roundoff, while allowing us to distribute the large atomic graph over the memory of multiple CPU or GPU nodes.

DeePMD force field

The atomic structures used for the large graphene calculation are relaxed with a Deep Potential force field. This model replaces repeated first-principles calculations by writing the total energy as a sum of atomic contributions determined by each atom's local environment [43],

$$E_\vartheta(\{\boldsymbol{R}_i\}) = \sum_i \varepsilon_\vartheta(\mathcal{N}_i), \qquad \boldsymbol{F}_i = -\frac{\partial E_\vartheta}{\partial \boldsymbol{R}_i}. \tag{36}$$

Here ϑ denotes the learned model parameters, $\mathcal{N}_i$ contains the neighbours of atom i within a finite cutoff, and the same neural network ε_ϑ is used for every carbon atom. The resulting cost grows approximately in proportion to the number of atoms. The force field is used only for this structural step; the subsequent electronic observables are obtained from the DeePTB operators with RSATB.

The model was trained with the DP-GEN active-learning workflow [44]. Four independently initialized models explored AA and AB stackings in bilayer cells, and AAA, ABA and ABC stacking of trilayers of several sizes, relative translations between the layers and triangular graphene cells. Constant-temperature, constant-pressure molecular dynamics was performed at 25, 300 and 600 K and at pressures of 0, 1, 10 and 100 bar. Configurations with largest force deviation between 0.18 and 0.45 eV Å^{-1} were labelled by density-functional theory. These reference calculations used the PBE functional with vdw-D3 correction, optimized norm-conserving Vanderbilt pseudopotentials and numerical atomic orbitals in ABACUS [42, 45]. The explore–label–retrain cycle was continued until it found no further high-disagreement structures in the sampled temperature and pressure range. The resulting force field after the active learning gives energy root-mean-square errors are 2.187 and 0.16 meV per atom for the training and validation sets, and the corresponding force errors are 0.095 and 0.027 eV Å^{-1} per Cartesian component, as displayed in Figure. 6. Similar to the DeePTB, validation structures include first-principles molecular-dynamics trajectories of moiré graphene at twist angles excluded from training, providing a transfer test from the small training cells to moiré geometries.

Scaling benchmark protocol

The scaling benchmark uses a commensurate twisted-bilayer-graphene cell at $\theta \simeq 1.08^\circ$, containing 11,164 atoms. We expand this cell from 1×1 to 10×9 in approximately twofold size increments, producing systems of 11,164 to 1,004,760 atoms. DeePTB predicts H in a C 2s2p1d basis with 13 orbitals per atom, giving 1.31×10^7 orbitals. All calculations sample the Γ point and estimate the trace with four random vectors. The DOS is evaluated by the kernel polynomial method with a Jackson kernel and 30-meV resolution. A fixed Chebyshev window of $[-22.5, 233.0]$ eV gives 31,503 moments for every system.

Calculations run on Trillium CPU nodes of Compute Canada, each containing two AMD EPYC 9655 processors with 192 cores and 768 GB of memory. Each node uses eight MPI ranks and 24 OpenMP threads per rank. The weak-scaling series fixes the load at 89,312 atoms per node and uses one, two, four and eight nodes, spanning 89,312 to 714,496 atoms. Two fixed-resources runs use one node for 11,164–89,312 atoms and eight nodes for 89,312–1,004,760 atoms. Peak memory is summed across nodes and fitted as $M = aN_{\text{node}} + bN_{\text{atom}}$ to separate the per-node runtime allocation from storage that grows with system size.

Data availability

The full data in this study is available upon request, and will be publicly available upon publication. The data includes frozen source code version used to run the calculation, the input configuration and data computed from the benchmark, scaling and application.

Code availability

RSATB code is available as an open-source package from Github under https://github.com/DeepElectron/rsatb/. The two-centre integral package Op2c and variable block sparse matrix backend are also available openly on Github via https://github.com/floatingCatty/op2c and https://github.com/DeepElectron/vbcsr.

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Author contributions

Z.Z. and H.G. conceived the idea and designed this work. Z.Z. developed the main framework of the RSATB package. L.R. and Z.Z. developed the real space properties of the package. Q.G. and Z.Z. developed the DeePTB parallel extension. Z.Z. and H.G. wrote the manuscript. Z.Z. L.R. Q.G. and H.G. discussed and revised the manuscript.

Competing interests

The authors declare no competing interests.

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